附錄

階乘

n	$n!$	$1/n!$
1	1	1
2	2	0.5
3	6	0.16666666666666666666
4	24	0.04166666666666666666
5	120	0.00833333333333333333
6	720	0.00138888888888888888
7	5040	0.00019841269841269841
8	40320	0.00002480158730158730
9	362880	0.00000275573192239858
10	3628800	0.00000027557319223985
11	39916800	0.00000002505210838544
12	479001600	0.00000000208767569878
13	6227020800	0.00000000016059043836
14	87178291200	0.00000000001147074559
15	1307674368000	0.00000000000076471638
16	20922789888000	0.00000000000004779477
17	355687428096000	0.00000000000000281145
18	6402373705728000	0.00000000000000015619
19	121645100408832000	0.82206×10^{-17}
20	2432902008176640000	0.41103×10^{-18}

n	a	c
25	1.5511	25
30	2.6525	32
35	1.0333	40
40	8.1592	47
45	1.1962	56
50	3.0414	64
60	8.3210	81
70	1.1979	100
80	7.1569	118
90	1.4857	138
100	9.3326	157
120	6.6895	198
140	1.3462	241
160	4.7147	284
180	2.0090	329
200	7.8866	374
220	2.2839	421
240	4.0679	468

二項係數

$$\binom{N}{n}=\frac{N!}{n!\,(N-n)!}$$

n \ N	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2		1	3	6	10	15	21	28	36	45	55	66	78	91	105	120	136	153	171	190
3			1	4	10	20	35	56	84	120	165	220	286	364	455	560	680	816	969	1140
4				1	5	15	35	70	126	210	330	495	715	1001	1365	1820	2380	3060	3876	4845
5					1	6	21	56	126	252	462	792	1287	2002	3003	4368	6188	8568	11628	15504
6						1	7	28	84	210	462	924	1716	3003	5005	8008	12376	18564	27132	38760
7							1	8	36	120	330	792	1716	3432	6435	11440	19448	31824	50388	77520
8								1	9	45	165	495	1287	3003	6435	12870	24310	43758	75582	125970
9									1	10	55	220	715	2002	5005	11440	24310	48620	92378	167960
10										1	11	66	286	1001	3003	8008	19448	43758	92378	184756
11											1	12	78	364	1365	4368	12376	31824	75582	167960
12												1	13	91	455	1820	6188	18564	50388	125970
13													1	14	105	560	2380	8568	27132	77520
14														1	15	120	680	3060	11628	38760
15															1	16	136	816	3876	15504
16																1	17	153	969	4845
17																	1	18	171	1140
18																		1	19	190
19																			1	20
20																				1